From the Quantum Universe to Quantum Technologies
Spring 2025: Week 10

Quantum Computing
Implementing Circuits

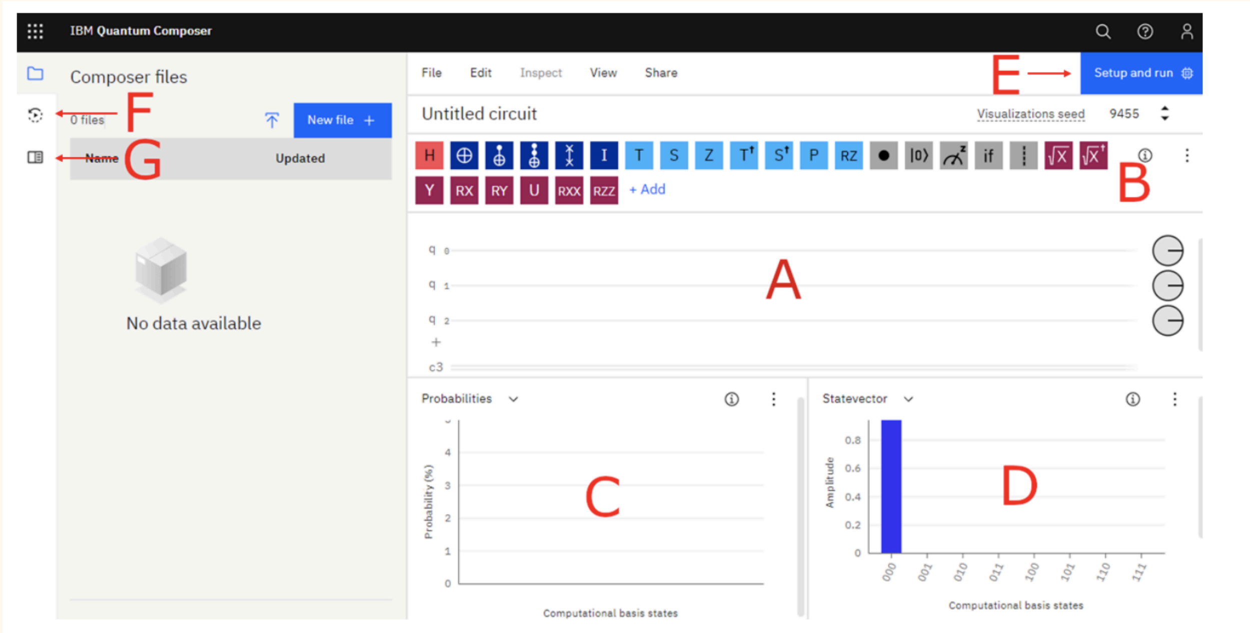
—

Getting Set Up

Introduction: IBMs Code

Launch:
https://quantum.ibm.com/composer/files/new?initial=N4IgdghgtgpiBcICqYAuBLVAbGATABAMboBOhArpiADQgCOEAzlAiAPIAKAogHICKAQQDKAWXwAmAHQAGANwAdMOjCEs5XDHzz6MLOgBGARknLC2hWEV0SMAOb46AbQAsAXQuEb9wi-eKAFg6Ohn5gNCAajJ7oAA4YAPZhiCAAvkA

Multiple windows should appear. Please close the window labeled ’OpenQASM 2.0’ or ’Qiskit’. Change the window labeled “Q-sphere” to a “State vector” window by clicking on “Q-sphere” and selecting “State vector”.

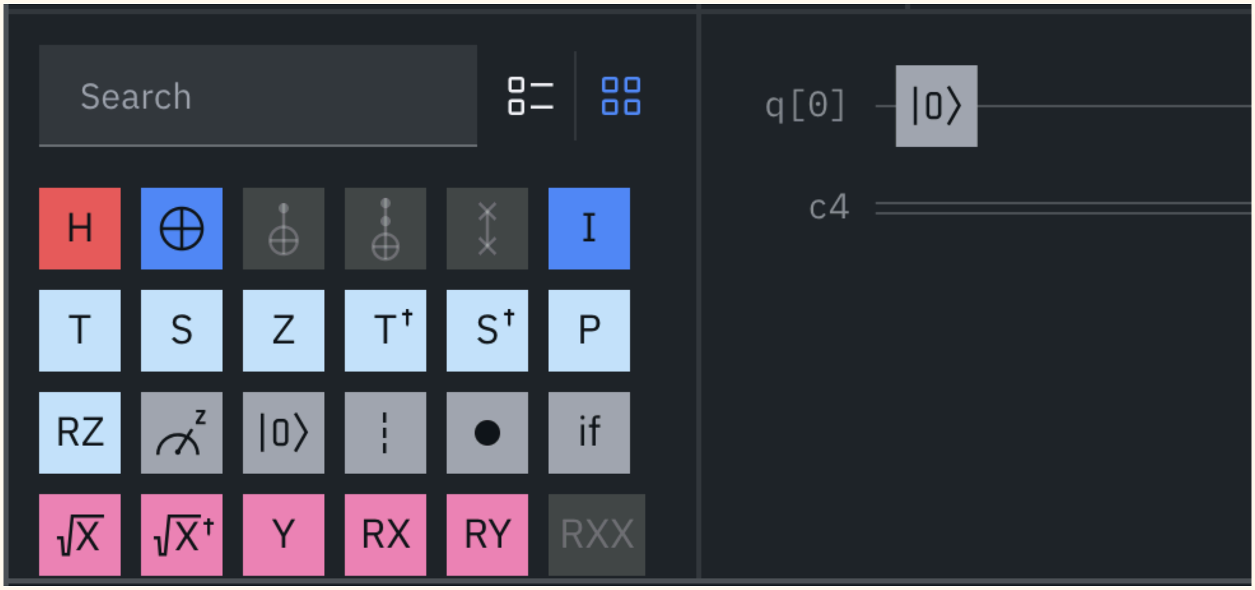


In the image we see the Quantum Composer user interface.

- A: The quantum circuit in which gates will be placed
- B: Menu with gates available in the quantum composer
- C: The simulated results from running the quantum circuit
- D: The simulated state vector of all the qubits after running the quantum circuit
- E: Button to run the circuit on actual IBM quantum hardware
- F: Job status and results
- Documentation and tutorials by IBM.

The representation of qubits and gates in the image is usually referred to as the quantum circuit. The qubits are represented by the horizontal lines labeled q[0], q[1] and so on. The line labeled c4 is a classical bit. In this assignment, we will only perform single-qubit operations on q[0], so please delete all other qubits by clicking on them, and then clicking the trash can icon.

The Quantum Composer will automatically simulate the quantum circuit every time the circuit is changed. The results of this simulation are visible in windows C and D. Window D shows the final state vector of the qubits after all the gates have been applied. Window C shows the probability of getting specific outcomes in the our usual basis {|0⟩, |1⟩} when the qubits are measured. Initially, the qubit is in the |0⟩ state.

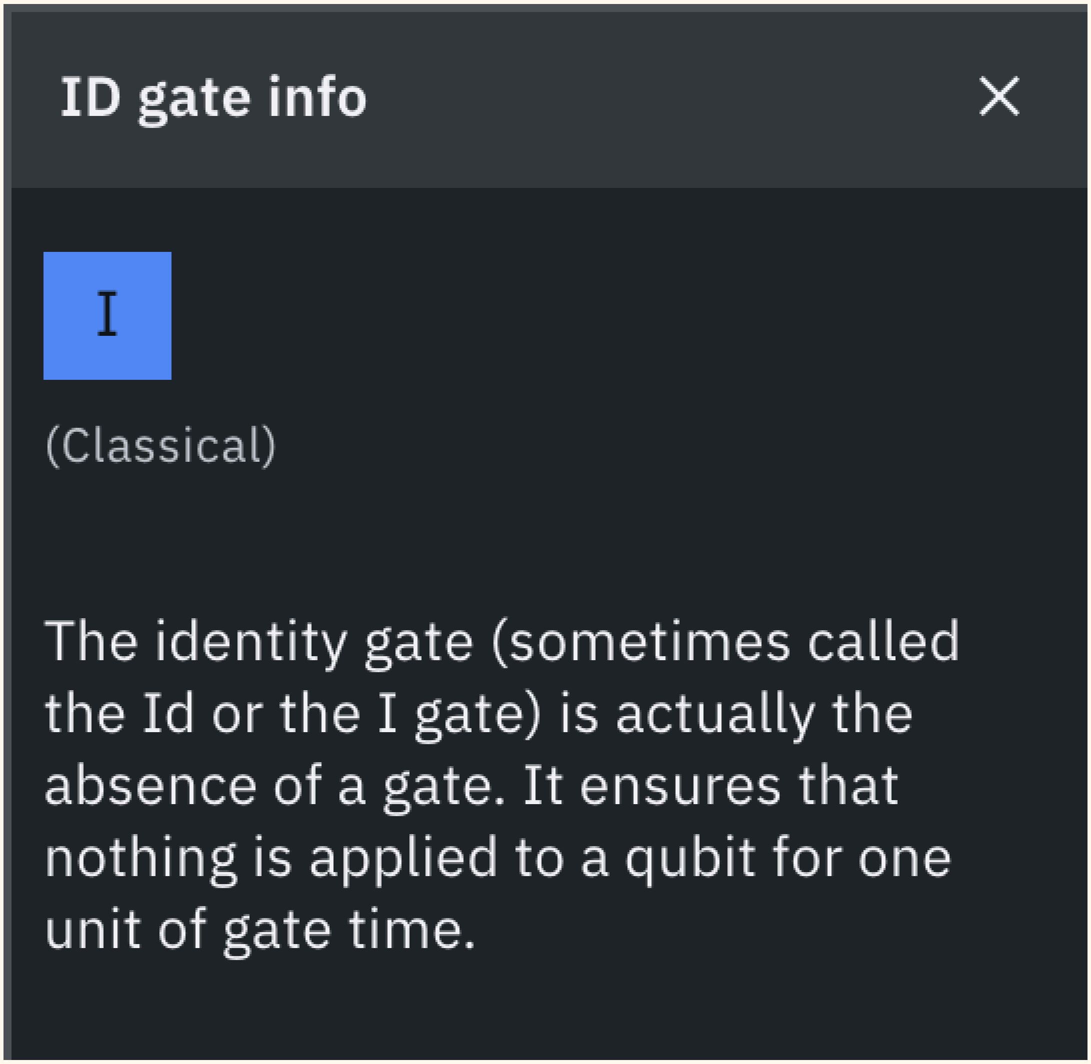


Guided Questions

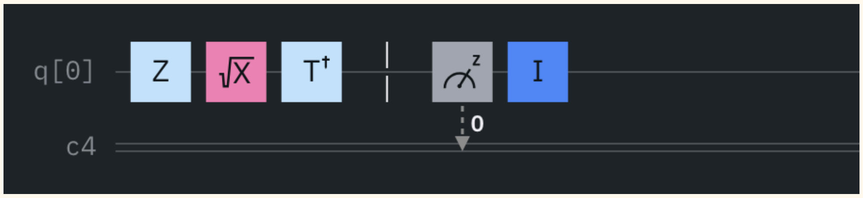
Preparing the Qubit

1. Apply the |0> to the qubit to put it into a state. What is its state? What is the probability of this state?

2. Apply the H gate. What did this do to the state and probabilities? Does this make sense with what we did 3 weeks ago?
3. **Right clicking on the gate icons will give us information about them.** Pick two gates you do not know and explain what they are called and what they do.

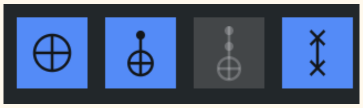


4. Use any combination of the gates: See what happens to the probability distribution. Try at least 3 different circuits. Write down: Which gates you used, Your expected outcome, and What you observed. Add a second qubit and see if anything changes.



Section 2: A Challenge

1. Build a circuit that produces 75% chance of |0⟩ and 25% of |1⟩. (Hint: Try adding a Ry gate with an angle that affects the state amplitude.)
    a. Follow-up: What angle gives you 50/50? What about 90/10?
2. Use a combination of gates that flips a qubit to |1⟩ and then back to |0⟩. (This is doable with only X and H gates)
3. Create a 2-qubit circuit that ends in: |00⟩ with 100% probability. Look at the info section for these pictured operations, which may be helpful:



    a. Then modify it to produce |11⟩ with 100%. (Hint: Think about where to place X gates)
4. Make a 2-qubit system that, when measured:
    a. Shows |00⟩ and |11⟩ each with 50% probability. (Hint: Start with an H gate and then use a CNOT.)
    b. Follow-up: What happens if you swap control/target?
5. Build a circuit with at least 3 gates, but when you run it, the output is always |0⟩ — just like if you ran no gates at all.

Section 3: Conceptual Questions

1. How does a quantum gate like the Hadamard differ from a classical logic gate?
    a. Compare how they affect information and how reversible they are.
    b. Why is reversibility important in quantum computing?